

ES-205

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Chapter: 01

Linear Equations in Linear Algebra

1.1 System of Linear Equations

$$y = a_1x_1 + a_2x_2 + \dots + a_nx_n \quad \text{Linear equation}$$

Consistent vs Inconsistent solution

↓
When it has
a solution

$$x_1 + 2x_2 + 3x_3 = 1$$

$$4x_1 + 5x_2 + 6x_3 = 2$$

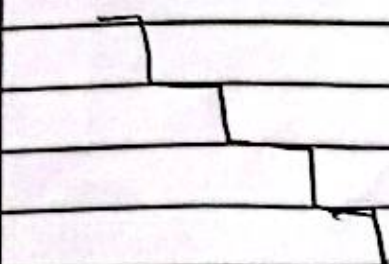
$$7x_1 + 8x_2 + 10x_3 = 3$$

$$\begin{bmatrix} 1 & 2 & 3 & : & 1 \\ 4 & 5 & 6 & : & 2 \\ 7 & 8 & 10 & : & 3 \end{bmatrix}$$

→ Augmented Matrix

$[A : b]$

Echelon



Why do these ops
NOT alter the solution?
↳ Because eqs are
a balance and since
ops are applied to
both.

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- ↳ Echelon form can be different for a given matrix
- ↳ Reduced Echelon form is one & only one for a given matrix

$$\begin{bmatrix} 1 & 2 & 3 & : & 1 \\ 4 & 5 & 6 & : & 2 \\ 7 & 8 & 10 & : & 3 \end{bmatrix} \quad \begin{array}{l} 4R_1 - R_2 \rightarrow R_2 \\ 7R_1 - R_3 \rightarrow R_3 \end{array}$$

$$2 \quad \begin{bmatrix} 1 & 2 & 3 & : & 1 \\ 0 & 3 & 6 & : & 2 \\ 0 & 6 & 11 & : & 4 \end{bmatrix} \quad 2R_2 - R_3 \rightarrow R_3$$

$$2 \quad \begin{bmatrix} 1 & 2 & 3 & : & 1 \\ 0 & 3 & 6 & : & 2 \\ 0 & 0 & 1 & : & 0 \end{bmatrix}$$

$$x_3 = 0$$

$$3x_2 + 6x_3 = 2$$

$$x_2 = \frac{2}{3}$$

$$x_1 + 2x_2 + 3x_3 = 1$$

$$x_1 = -\frac{1}{3}$$

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$$\begin{bmatrix} 1 & 2 & 3 & : & 1 \\ 4 & 5 & 6 & : & 2 \\ 7 & 8 & 9 & : & 3 \end{bmatrix}$$

 $x_1 \quad x_2 \quad x_3$

$$\begin{bmatrix} 1 & 2 & 3 & : & 1 \\ 0 & 3 & 6 & : & 2 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

As there is no leading
value at position x_3
we take it as free variable

 $x_3 = \text{free variable}$

$$3x_2 + 6x_3 = 2$$

$$x_1 + 2x_2 + 3x_3 = 1$$

$$x_2 = \frac{2}{3} - 2x_3$$

$$x_1 + 2\left(-2x_3 + \frac{2}{3}\right) + 3x_3 = 1$$

$$x_1 = x_3 - \frac{1}{3}$$

$$= \begin{bmatrix} x_3 - \frac{1}{3} \\ -2x_3 + \frac{2}{3} \\ x_3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & & & \frac{1}{3} \\ x_3 & -2 & + & \frac{2}{3} \\ & 1 & & 0 \end{bmatrix}$$

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Theorem 2.

↳ A linear system is consistent if and only if the rightmost column of the augmented matrix is not a pivot column.

Example,

$$x_1 + 2x_2 = 3$$

$$4x_1 + 5x_2 = 6$$

$$Ax = b$$

$$\begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$$

$$\begin{array}{cc|c} x_1 & x_2 & b \\ 1 & 2 & 3 \\ 4 & 5 & 6 \end{array}$$

$$4x_1 - x_2 \rightarrow t_2$$

$$\begin{array}{cc|c} \text{pivot} \uparrow & & \text{below} \\ \text{as } 0 & & \\ \hline 1 & 2 & 3 \\ 0 & 3 & 6 \end{array}$$

↓
pivot as
nothing
below

$$x_2 = 2$$

$$x_1 = -1$$

$$7x_1 + 8x_2 = 10$$

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$$\begin{array}{l} 1 \quad 2 : 3 \\ 4 \quad 5 : 6 \\ 7 \quad 8 : 10 \end{array} \quad \begin{array}{l} \\ 4x_1 - x_2 \rightarrow x_2 \\ 7x_1 - x_3 \rightarrow x_3 \end{array}$$

$$\sim \begin{array}{l} 1 \quad 2 : 3 \\ 0 \quad 3 : 6 \\ 0 \quad 6 : 11 \end{array} \quad \begin{array}{l} \\ 2x_2 - x_3 \rightarrow x_3 \end{array}$$

$$\sim \begin{array}{l} 1 \quad 2 : 3 \\ 0 \quad 3 : 6 \\ 0 \quad 0 : 1 \end{array}$$

↳ pivot here. so

↳ No solution i.e. Inconsistent

Exercise 1.3

↳ Vectors in $\mathbb{R}^2$, $\mathbb{R}^3$ and $\mathbb{R}^n$

↳ Linear combinations of Vectors and their applications

$$a_1 \underline{v}_1 + a_2 \underline{v}_2 = \underline{u}$$

$$a_1 = 3, a_2 = -2$$

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$$Ax = b$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b \\ \\ \end{bmatrix}$$

↳ multiplied by x_1 always when matrix mul

$$x_1 v_1 + x_2 v_2 + x_3 v_3 = b$$

Example,

$$C = \begin{bmatrix} 0.4 \\ 0.3 \\ 0.15 \end{bmatrix} \quad b = \begin{bmatrix} 0.45 \\ 0.25 \\ 0.15 \end{bmatrix}$$

$$x_1 b + x_2 C$$

Span

↳ A set of vectors

$$\text{span}\{v_1\} = c v_1$$

↳ All scalars

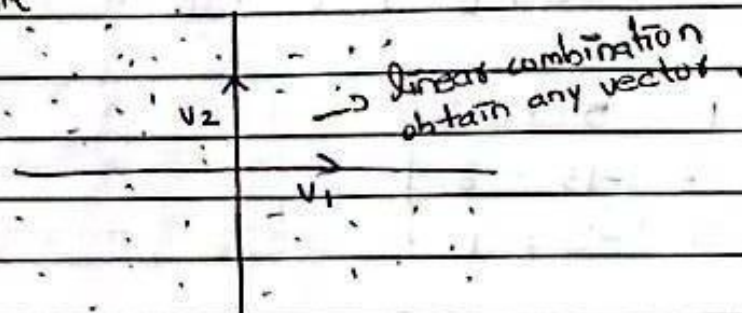
→ scale this up & down (-1 to get a line which is its span

$$\text{span}\{v_1, v_2\}$$

$$v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

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 $C_1 V_1 + C_2 V_2$ } complete two d space
Any $\mathbb{R}$ 
 When can we say b is in span $\{v_1, v_2, \dots, v_n\}$

$$C_1 V_1 + C_2 V_2 + \dots + C_n V_n = \underline{b}$$

$$\begin{bmatrix} V_1 & V_2 & \dots & V_n \end{bmatrix} \begin{bmatrix} C_1 \\ \vdots \\ C_n \end{bmatrix} = \underline{b}$$

 c) When this has a solution / consistent, then we can say b is in span $\{v_1, v_2, \dots, v_n\}$

Example,

$$\underline{a}_1 = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$$

$$\underline{a}_2 = \begin{bmatrix} 5 \\ -13 \\ -3 \end{bmatrix}$$

$$\underline{b} = \begin{bmatrix} -3 \\ 8 \\ 1 \end{bmatrix}$$

 is $\underline{b}$ in span $\{\underline{a}_1, \underline{a}_2\}$

$$C_1 \underline{a}_1 + C_2 \underline{a}_2 = \underline{b}$$

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$$\begin{bmatrix} 1 & 5 \\ -2 & -13 \\ 3 & -3 \end{bmatrix} \underline{x} = \begin{bmatrix} -3 \\ 8 \\ 1 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} 1 & 5 & -3 \\ -2 & -13 & 8 \\ 3 & -3 & 1 \end{array} \right]$$

$$\sim \left[\begin{array}{cc|c} 1 & 5 & -3 \\ 0 & -3 & 2 \\ 0 & 0 & -2 \end{array} \right]$$

∴ No solution i.e. inconsistent

$\underline{b}$ is NOT in $\text{span}\{\underline{a}_1, \underline{a}_2\}$

Exercise 1.4:

$$\underset{m \times n}{A} \underline{x} = \underset{n \times 1}{\underline{b}}$$

$$= \begin{bmatrix} | & | & \dots & | \\ \underline{a}_1 & \underline{a}_2 & \dots & \underline{a}_n \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \underline{b}$$

$$(c) \text{span}\{\underline{a}_1, \underline{a}_2, \dots, \underline{a}_n\}$$

$$(b) x_1 \underline{a}_1 + x_2 \underline{a}_2 + \dots + \underline{a}_n x_n = \underline{b}$$

$$[A : \underline{b}]$$

∴ $\mathbb{R}^m$

we can obtain every point in the $\mathbb{R}^m$

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Theorem 4

$$\underset{m \times n}{A} \underset{n \times 1}{x} = \underset{m \times 1}{b} \in \mathbb{R}^{m \rightarrow m\text{-dimensional space}}$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 : b_1 \\ 4 & 5 & 6 : b_2 \\ 7 & 8 & 9 : b_3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 : b_1 \\ 0 & 3 & 6 : 4b_1 - b_2 \\ 0 & 0 & 0 : (8b_1 - 2b_2) - 7b_1 + b_2 \end{bmatrix}$$

fails statement (d) $\hookrightarrow$ if one fails, rest also fails
 $\hookrightarrow$ 0, for solution to exist. Failing statement (a)

$$\begin{bmatrix} 1 & 2 & 3 : b_1 \\ 4 & 5 & 6 : b_2 \\ 7 & 8 & 10 : b_3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 : b_1 \\ 0 & 0 & 6 : 4b_1 - b_2 \\ 0 & 0 & 1 : (8b_1 - 2b_2) - 7b_1 - b_2 \end{bmatrix}$$

$\hookrightarrow$ Satisfies (d)

$\hookrightarrow$ Satisfies (a) & all others.

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Exercise 1.5

Homogeneous

$$\hookrightarrow Ax = b$$

$$b = 0$$

Non-Homogeneous:

$$b \neq 0$$

$$Ax = b$$

Solution set of H.S - $\boxed{?}$ - solution set of N.H.S
 $\hookrightarrow$ Trial solution for H.S is $x = 0$

Example,

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 & 3 & : & 0 \\ 4 & 5 & 6 & : & 0 \\ 7 & 8 & 9 & : & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 2 & 3 & : & 0 \\ 0 & 3 & 6 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

 $\hookrightarrow$ Take as F.V

$$x_3 \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

 $\hookrightarrow$ Geometrically, it is a vector (line)

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Non-homogeneous System:

$$\begin{bmatrix} 1 & 2 & 3 & 1 \\ 4 & 3 & 6 & 2 \\ 7 & 8 & 9 & 3 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 2 & 3 & 1 \\ 0 & 3 & 6 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$= x_3 \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} + \begin{bmatrix} -1/3 \\ 2/3 \\ 0 \end{bmatrix}$$

∴ Solution set of NHS = solution set of H.S
+ Translation vector

Ex 1.6

P_c = Total price of coal's output

P_e = Total price of electric output

P_s = Total Price of steel output

Total earning - Total expenditure
of a sector of a sector

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Dependency matrix
↓
where output goes

	C	E	S	
C	0	0.4	0.6	→ what each sector wants
E	0.4	0.5	0.2	
S	0.6	0.1	0.2	

=1

For coal:

$$P_C = 0.4 P_E + 0.6 P_S$$

For Electric,

$$P_E = 0.6 P_C + 0.1 P_E + 0.2 P_S$$

For Steel,

$$P_S = 0.4 P_C + 0.5 P_E + 0.2 P_S$$

$$Ax = b$$

$$P_C - 0.4 P_E - 0.6 P_S = 0$$

$$-0.6 P_C + 0.9 P_E + 0.2 P_S = 0$$

$$-0.4 P_C - 0.5 P_E + 0.8 P_S = 0$$

$$\begin{bmatrix} 1 & -0.4 & -0.6 \\ -0.6 & 0.9 & -0.2 \\ -0.4 & -0.5 & 0.8 \end{bmatrix} \begin{bmatrix} P_C \\ P_E \\ P_S \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & -0.94 & : & 0 \\ 0 & 1 & -0.85 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

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$$P_c - 0.94 p_s = 0$$

Determined $n \times n$

$$P_c = 0.94 p_s$$

Undetermined $n \times m$ 4×3

$$P_c = 0.85 p_s$$

→ infinitely

$$\underline{P} = \begin{bmatrix} 0.94 p_s \\ 0.85 p_s \\ p_s \end{bmatrix}$$

Over determined $m \times n$ 3×4

→ No solution

$$= p_s \begin{bmatrix} 0.94 \\ 0.85 \\ 1 \end{bmatrix}$$

other sector's Price
is this times

Balancing chemical eqs

$$\begin{bmatrix} C \\ H \\ O \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 8 \\ 0 \end{bmatrix} x_1 + \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} x_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} x_3 + \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix} x_4$$

Ex 1.7 linear independence

$$x_1 v_1 + x_2 v_2 + \dots + x_p v_p = \underline{0} \rightarrow \text{only trivial solution}$$

Assume,

we have non trivial solution $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

$$\underline{v_1} + \underline{v_2} = \underline{0}$$

$$\underline{v_1} = -\underline{v_2}$$

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Determine if $\{v_1, v_2, \dots, v_p\}$ is a LI set?

$$x_1 v_1 + x_2 v_2 + \dots + x_p v_p = 0$$

$$\begin{bmatrix} 1 & 4 & 2 & 0 \\ 2 & 5 & 1 & 0 \\ 3 & 6 & 0 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$x_2 + x_3 = 0$$

$$x_1 - 2x_3 = 0$$

$$x_2 = -x_3$$

$$x_1 = 2x_3$$

$$x = \begin{bmatrix} 2x_3 \\ -x_3 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} x_3$$

Linearly dependent

$\{v\}$ is LI if $v \neq 0$

$$x_1 v_1 = 0, \quad v_1 = 0$$

for any x_1 ,

this eq is satisfied

$$\begin{pmatrix} -1 \\ 2 \\ 3 \\ 5 \end{pmatrix} \quad \begin{pmatrix} 1 \\ -2 \\ -3 \\ 5 \end{pmatrix}$$

$$\begin{bmatrix} x_1 & x_2 & x_3 & x_4 \\ 0 & 1 & 2 & 10 \\ 2 & 0 & 2 & 11 \\ 3 & 6 & 2 & 12 \end{bmatrix} \begin{matrix} 0 \\ 0 \\ 0 \\ 0 \end{matrix}$$

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$$x_1 \underline{v_1} = 0, \underline{v_1} \neq 0$$

Now, we have only trivial solution

For two vectors,

geometrically have same direction or opposite

$$Ax = 0$$

only trivial solution $\rightarrow$ Linearly independent

Any non-trivial solution $\rightarrow$ Linearly dependent

$\{v_1, v_2, v_3\} \rightarrow$ linearly dependent set

$v_1 = 2v_2$ but no relation of v_3

$$\text{span}\{u, v\} = cu + bv = w$$

$\{u, v, w\} \rightarrow$ linearly dependent set
w has relation with other

$$\begin{bmatrix} A \\ \vdots \end{bmatrix}_{3 \times 3} \rightarrow \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\{u, v, w, 0\}$$

$$\text{sol} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 3 \end{pmatrix}$$

$$x_1 \underline{u} + x_2 \underline{v} + x_3 \underline{w} + x_4 \underline{0} = 0$$

$\leftarrow$ satisfies but is not trivial

GALAXY

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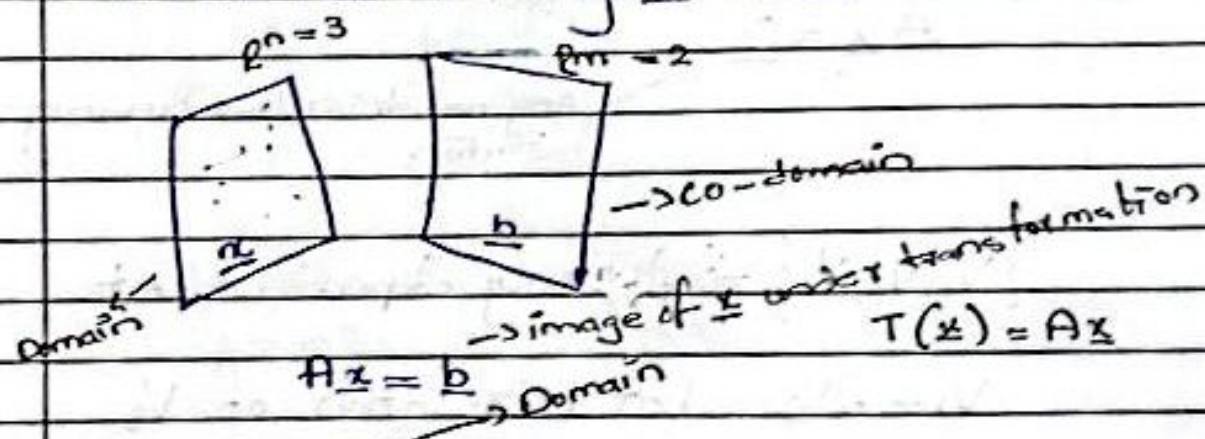
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Exercise 1.8

$$Ax = b \rightarrow \text{final vector}$$

↓
initial vector

↳ A is transforming x to b



$$T(x): \mathbb{R}^n \rightarrow \mathbb{R}^m \rightarrow \text{co-domain}$$

Exp:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

↳ Any vector in 3D space (Domain)

$$\begin{bmatrix} x_1 \\ x_2 \\ 0 \end{bmatrix}$$

belongs to $\mathbb{R}^3$ but 3rd is 0

↳ Any vector in 3D space covering 2D space



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Let $A = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix}$, $u = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, $b = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}$, $c = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$

and $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T(x) = Ax$

a) Find image of u under T ?

$$Au$$

b) Find x in $\mathbb{R}^2$ whose image under T is b .

$$Ax = b$$

$$[A : b]$$

$$x = \begin{bmatrix} 1.5 \\ -0.5 \end{bmatrix}$$

c) Is it more than one x , whose image under T is b ?

Since we have a unique solution, so no.

d) Determine if c is in the range of transformation T ?

$$Ax = c \quad \sim \quad \begin{bmatrix} 1 & 3 & : & 3 \\ 0 & 1 & : & 2 \\ 0 & 0 & : & 1-35 \end{bmatrix}$$

$$[A : c]$$

is inconsistent

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Linear Transformation:

↳ Superposition principle

$$T(\underline{au + bv}) = aT(\underline{u}) + bT(\underline{v})$$

w

↳ LT

$$T(\underline{w}) = T(\underline{au + bv})$$

$$= A \underline{w}$$

$$= A(\underline{au + bv})$$

$$= aA\underline{u} + bA\underline{v}$$

$$= aT(\underline{u}) + bT(\underline{v})$$

∴ If transformation of whole system is equivalent to transformation on individual vectors, then it is linear.

[Property of linearity]

$$f(x+w) = f(x) + f(w)$$

Every Matrix Transformation is Linear Transformation

$$T(\underline{x}) = A\underline{x} + \underline{t}$$

$$\underline{t} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$T(\underline{w}) = A\underline{w} + \underline{t}$$

$$= A(\underline{au + bv}) + \underline{t}$$

$$= aA\underline{u} + \underline{t} + bA\underline{v} = T(\underline{au}) + bA\underline{v}$$

$$T(\underline{au}) = aA\underline{u} + \underline{t}$$

$$\neq aT(\underline{u}) + bT(\underline{v})$$

$$T(\underline{bv}) = aA\underline{v} + \underline{t}$$

$$T(v+w) = T$$

$$\begin{bmatrix} 4x_1 - 2x_2 \\ 3x_2 \end{bmatrix} = 2_1 \begin{bmatrix} 4 \\ 0 \end{bmatrix} + 2_2 \begin{bmatrix} -2 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} 4 & -2 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

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$$T(x) \rightarrow \begin{bmatrix} x_1^2 \\ x_2^2 \end{bmatrix}$$

$$T(2_1 v_1 + 2_2 v_2) = T(x_{N1})$$

$$= T(x_2 v_2)$$

$$2_1 T(v_1) + 2_2 T(v_2)$$

$$x_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, x_2 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$T(x_1 + x_2) \neq T(x_1) + T(x_2)$$

Now,

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$$

$$T(x_1 + x_2) = A(x_1 + x_2)$$

$$= \begin{bmatrix} 4 \\ 4 \end{bmatrix}$$

$$T(x_1) + T(x_2) = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$= \begin{bmatrix} 4 \\ 4 \end{bmatrix}$$

Example 01:

Say $z \in \mathbb{R}^2$

$$T(z) = z_1 T(e_1) + z_2 T(e_2)$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\underbrace{[T(e_1) \ T(e_2)]}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_x$$

$$z_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + z_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

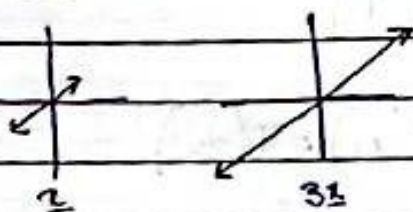
$$z_1 \underline{e_1} + z_2 \underline{e_2}$$

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Example, 02:

Find the standard matrix for A dilation transformation?
 Given by $3x$ in $\mathbb{R}^2$



$$T(\underline{x}) = A\underline{x} = 3\underline{x}$$

$$A = [T(e_1) \ T(e_2)]$$

$$e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$T(e_1) = \begin{bmatrix} 3 \\ 0 \end{bmatrix}, T(e_2) = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$$

$$A\underline{x} = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 6 \\ 3 \end{bmatrix}$$

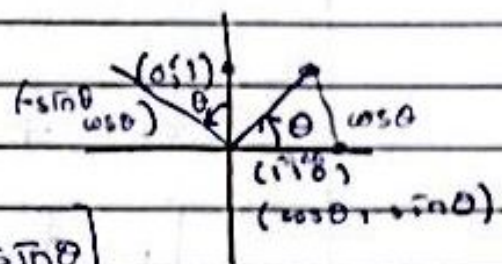
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Example 03:

"rotation" which rotate each vector by θ .

$$\mathbb{R}^2 \rightarrow A = [T(e_1) \ T(e_2)]$$



$$T(e_1) = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}, \quad T(e_2) = \begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix}$$

We can do this since len of vectors, is $1, \sqrt{(\cos \theta)^2 + (\sin \theta)^2} = 1$

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

onto is when ω -domain = range

$$Ax = b$$

$$A \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$\mathbb{R}^k$ If $\mathbb{R}^k$ covers entire $\mathbb{R}^2$, so it is onto $\mathbb{R}^2$

$$Ax = b$$

unique then one-to-one sol

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Ex 1.10

Cambridge Dict:

$$x_1 m + x_2 f + x_3 w = 0$$

$$\begin{bmatrix} m & f & w \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 33 \\ 45 \\ 3 \end{bmatrix}$$

Dynamic System

updates its state

x_k = state of system at instance k .

$$x_{k+1} = A x_k$$

↓
updated version
of x_k .

$$x = \begin{bmatrix} c \\ s \end{bmatrix}$$

$$x_0 = \begin{bmatrix} c_0 \\ s_0 \end{bmatrix}, x_1 = \begin{bmatrix} c_1 \\ s_1 \end{bmatrix} = \begin{bmatrix} 0.95 & 0.03 \\ 0.05 & 0.97 \end{bmatrix} \begin{bmatrix} c_0 \\ s_0 \end{bmatrix} \begin{matrix} \rightarrow \text{City} \\ \rightarrow \text{Suburb} \end{matrix}$$

$$x_1 = A x_0 = \begin{bmatrix} 0.95 & 0.03 \\ 0.05 & 0.97 \end{bmatrix} \begin{bmatrix} c_0 \\ s_0 \end{bmatrix}$$

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After 2 years;

After 100 years Date: / /

$$\underline{x}_2 = A \underline{x}_1$$

$$\underline{x}_{100} = A^{100} \underline{x}_0$$

$$= A A \underline{x}_0$$

$$= A^2 \underline{x}_0$$

all
matrices
 $n \times n$

Chapter 02:

Ex 2.1

ABx — composition of mappings

↳ x mapped by B

↳ result mapped by A

Transformation.

$$AB = A [b_1 \dots b_p]_{n \times p}$$

$$= [Ab_1 \dots Ab_p] \rightarrow \text{multiplying individual columns by } A \text{ is same as whole } AB$$

$$C_{ij} = \underbrace{[a_i]}_{\text{ith row of } A} \cdot \underbrace{[b_j]}_{\text{jth column of } B}$$

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$$\begin{bmatrix} A \end{bmatrix} \begin{bmatrix} B \end{bmatrix} \begin{bmatrix} C \end{bmatrix} \quad p \leq n$$

$n \times m \quad m \times m \quad n \times p$

Pattern matching

if $\rightarrow$ Magic Matrix

$$x^T M x = 0 \Rightarrow x \text{ is desired pattern}$$

else

NOT

$$M = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}$$

Data Scrubbing:

Swap Row C to match the pattern

$$\begin{matrix} A & C \\ \uparrow \\ \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \end{matrix}$$

$$AB = C$$

$\begin{bmatrix} \end{bmatrix} \begin{bmatrix} \end{bmatrix} = \begin{bmatrix} \end{bmatrix} \rightarrow$ Each column of C is obtained from linear combination of the columns of A

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$$c_1 = b_{11}a_1 + b_{21}a_2 + b_{31}a_3 + \dots + b_{n1}a_n$$

$$c_2 = b_{12}a_1 + b_{22}a_2 + \dots + b_{n2}a_n$$

if rows instead of columns, then linear combination of B.

$$(AB)^T = C^T$$

$$B^T A^T = C^T$$

rows of B are columns of B^T .

Ex 2.2

$$AA^{-1} = I_n$$

NOT invertible = singular matrix

Invertible = non singular matrix

$$\underline{x} = A^{-1}b$$

Ill conditioned matrix:

$100 \times 100 \rightarrow$ has inverse but due to some numbers changing it can become non-invertible (maybe due to number errors during program)

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Elementary Matrices:

① $\hookrightarrow$ Row exchange② $\hookrightarrow$ Row scaling③ $\hookrightarrow$ Row replacement

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

① Row Exchange

$$r_1 \leftrightarrow r_2 = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$$

$$E_1 A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$$

② Row scaling:

$$2r_1 \rightarrow r_1 \Rightarrow E_2 A$$

$$\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 3 & 4 \end{bmatrix}$$

③ Row replacement

$$3r_1 - r_2 \rightarrow r_2 = E_3 A$$

$$\begin{bmatrix} 1 & 0 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$$

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$$[A : I_n] \rightarrow [I_n : A^{-1}]$$

$$(\varepsilon_1 \dots \varepsilon_p) A = I_n$$

All these row operation are defining A^{-1} .

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$A^{-1} = \begin{array}{c|cc} 1 & 4 & -2 \\ (-2) & -3 & 1 \end{array} \begin{array}{l} \text{Adj } A \\ |A| \end{array}$$

$$= \begin{bmatrix} -2 & 1 \\ 3/2 & -1/2 \end{bmatrix} \xrightarrow{W} \varepsilon_1 = \begin{bmatrix} 1 & 0 \\ 3 & -1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix} \xrightarrow{1/2 \rightarrow r_2} \varepsilon_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1/2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \xrightarrow{-2r_2 + r_1 \rightarrow r_1} \varepsilon_3 = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

we have

$$A^{-1} = \varepsilon_3 \cdot \varepsilon_2 \cdot \varepsilon_1$$

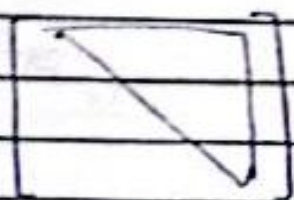
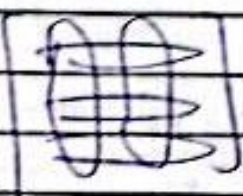
$$= \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1/2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 3 & -1 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 3/2 & -1/2 \end{bmatrix} \xrightarrow{W}$$

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$$[A \quad \underline{e}_i] = [I_n : \bar{a}_i^1]$$

Ex 2.4

→ Upper & lower
partition→ column/partitions
row

A_{11}	A_{12}	A_{13}
A_{21}	A_{22}	A_{23}

→ one partition

$$\begin{bmatrix} \overset{A_{11}}{1} & 2 & 5 \\ \underset{A_{21}}{3} & 4 & 6 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{matrix} \rightarrow \text{Same partition} \\ \text{size} \\ \text{can't do this} \end{matrix}$$

$B_{11} \quad B_{12}$

→ Breaking into partition helps save memory

$$\begin{bmatrix} 1 & 2 & 5 \\ 3 & 4 & 6 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 0 \end{bmatrix} \rightarrow \text{NOT compatible by partition}$$

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$$\left[\begin{array}{cc|c} 1 & 2 & 5 \\ 3 & 4 & 6 \end{array} \right] \left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 0 \end{array} \right]$$

$$\begin{bmatrix} A_{11} & A_{12} \end{bmatrix} \times \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$$

$$\begin{bmatrix} A_{11}B_{11} + B_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \end{bmatrix}$$

Purpose: Find Inverse

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{matrix} p \times p & p \times q \\ q \times p & q \times q \end{matrix} \quad \begin{matrix} \text{Block upper} \\ \text{triangular matrix} \end{matrix}$$

$p+q \times (p+q)$

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 1 \\ 3 & 4 & 1 & 1 \\ 0 & 0 & 5 & 6 \\ 0 & 0 & 7 & 8 \end{array} \right]$$

Now,

$$AA^{-1} = I_{(p+q) \times (p+q)}$$

$$\begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix} \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix} = \begin{bmatrix} I_p & 0 \\ 0 & I_q \end{bmatrix}$$

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$$A_{11}B_{21} + A_{12}B_{21} = I_p \quad \text{--- (2)}$$

$$A_{11}B_{12} + A_{12}B_{22} = 0 \quad \text{--- (3)}$$

$$A_{22}B_{21} = 0 \rightarrow B_{21} = 0$$

$$A_{22}B_{22} = I_p \rightarrow B_{22} = A_{22}^{-1}$$

Using (1) & (2)

$$A_{11}B_{11} = I_p$$

$$B_{11} = A_{11}^{-1}$$

(3) as:

$$A_{11}B_{12} = -A_{12}B_{22}$$

$$= -A_{12}A_{22}^{-1}$$

$$B_{12} = -A_{11}^{-1}A_{12}A_{22}^{-1}$$

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Ex 2.5

Matrix Factorization

$$x^2 - 1 = (x+1)(x-1)$$

$$14 = \underbrace{7}_{\text{factors}} \times \underbrace{2}_{\text{factors}}$$

breaking into

$$A = CD \quad \text{other techniques like POP or QR}$$

$$= LU$$

$$A \underline{x} = \underline{b}$$

$$LU \underline{x} = \underline{b}$$

(Unit
on diagonal)
 Lower
Upper (EF)
 triangular
matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 2 & 6 & 0 \end{bmatrix} \begin{bmatrix} 6 & -1 & 2 \\ 0 & 4 & 0 \\ 0 & 0 & 3 \end{bmatrix} \underline{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

L U

$$U \underline{x} = \underline{y}$$

$$L \underline{y} = \underline{b}$$

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$$\begin{array}{ccc|ccc} 1 & 0 & 0 & y_1 & & 1 \\ -4 & 1 & 0 & y_2 & = & 2 \\ 2 & 6 & 1 & y_3 & & 3 \end{array}$$

↓

Already in Echelon form.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$\varepsilon_p \dots \varepsilon_2 \varepsilon_1 A = U$$

$$A = (\varepsilon_p \dots \varepsilon_1)^{-1} U$$

$$= LU$$

$$\begin{bmatrix} 1 & 2 \\ 0 & -2 \end{bmatrix} \quad r_2 - 3r_1 \rightarrow r_2$$

$$\begin{bmatrix} 1 & 2 \\ 0 & -2 \end{bmatrix} \quad , \quad \varepsilon_1 = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}$$

U

$$\varepsilon_1 A = U$$

$$A = \varepsilon_1^{-1} U$$

$$L = \varepsilon_1^{-1} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$$

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$$[A]_{m \times n} = [L]_{m \times m} [U]_{m \times n}$$

Example

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \quad \begin{array}{l} r_1 - 4r_1 \rightarrow r_2 \\ r_1 - 7r_1 \rightarrow r_3 \end{array}$$

$$\sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{bmatrix} \quad r_3 - 2r_2 \rightarrow r_3$$

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -7 & 0 & 1 \end{bmatrix}, \quad E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{bmatrix}$$

Shest-cut:

Divide pivot column by its pivot

$$L = (E_3 E_2 E_1)^T$$

$$= E_1^{-1} E_2^{-1} E_3^{-1}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 7 & 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 7 & 2 & 1 \end{bmatrix}$$

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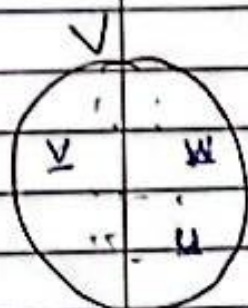
N-5

Chapter 04

Vector Spaces

↳ Helps dealing with infinite

Vectors Matrices Signals Images Text Polynomial



set of mathematical

principles

through which we can deal with
above

if satisfy right
then left is vector
space.

$$1. u, v \in V \wedge u + v \in V$$

$$2. 0 \in V$$

$$3. w \in V, cw \in V, c \in \mathbb{R}$$

Exp:

$$\mathbb{R}^n \text{ eg. } \mathbb{R}^2$$

$$1. \text{YES} \quad 2. \text{YES}$$

$$3. \text{NO, as } c > 0 \text{ for it to be } \mathbb{R} \wedge \text{condition reqs } c \text{ to be } \mathbb{R}$$

$$\begin{matrix} u & v_1 \\ \cdot & \\ \cdot & \\ \cdot & \end{matrix}$$

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Exp 02:

Matrices $\mathbb{R} \rightarrow$ set of all 2×2 matrices with $\mathbb{R}$ elements

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$$

Exp 03:

Polynomials $\mathbb{P}_n =$ set of all polynomials of degree n

$$\mathbb{P}_2 = \overset{\mathbb{R}}{a_0 + a_1x + a_2x^2}$$

$$1) P_1 = a_0 + a_1x + a_2x^2$$

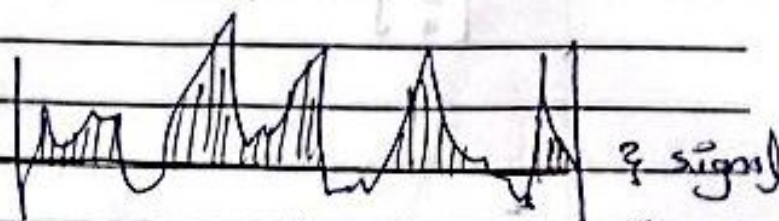
$$P_2 = b_0 + b_1x + b_2x^2$$

$$P_3 = a_0 + b_0 + (a_1 + b_1)x + (a_2 + b_2)x^2$$

$$2) 0 \in \mathbb{P}_2 \rightarrow 0 + 0x + 0x^2 = 0$$

$$3) c \underline{p} \in \mathbb{P}^n$$

Exp 04:



1) ✓ Defining new sequence will also be a seq of $\mathbb{R}$ in an interval

sequence of Real numbers defined in an interval.

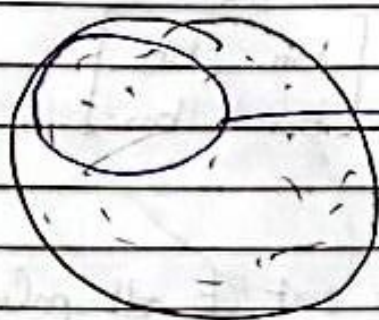
2) ✓ No amplitude

3) ✓

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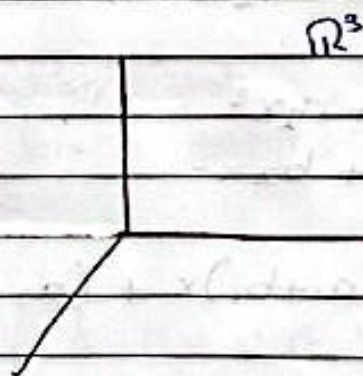
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Vector Subspace



→ A such a subset that observes the properties of a vector space.

is $\mathbb{R}^2$ a subspace of $\mathbb{R}^3$? NO!



is $\begin{bmatrix} x \\ y \\ 0 \end{bmatrix}$, where $x, y \in \mathbb{R}$ is a subspace of $\mathbb{R}^3$.

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Theorem 01.

$$\text{span} \{H = c_1 \underline{v}_1 + c_2 \underline{v}_2$$

$$1) \underline{0} \in H, c_1 = c_2 = 0 \quad \checkmark$$

$$2) h_1 = c_{11} \underline{v}_1 + c_{12} \underline{v}_2 \quad \checkmark$$

$$h_2 = c_{21} \underline{v}_1 + c_{22} \underline{v}_2$$

$$h_1 + h_2 = (c_{11} + c_{21}) \underline{v}_1 + (c_{12} + c_{22}) \underline{v}_2 \quad \checkmark$$

$$3) sh_1 \in H$$

↳ Spanning set is finite
Spanning set is infi

Ex 4.2.

$$Ax = b$$

$$[\begin{bmatrix} | & & | \end{bmatrix}]$$

Set of all of columns of A. (Not a sub-space)

But $\text{span} \{ " \}$ (A subspace)

↓
column space

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Row space = $\text{span} \{ \text{row}^T \text{ vectors of } A \}$

Null spaces

$$Ax = 0$$

Set of all x which satisfy above eq. = $\text{Nul } A$

1) $0 \in \text{Nul } A$ as trivial sol always exists ✓

2) $Ax_1 = 0, Ax_2 = 0$ ✓

$$A(x_1 + x_2) = 0$$

3) $A(cx_1) = 0$ ✓

Example:

↳ A linear system given by:

$$x_1 + 2x_2 + 3x_3 = 0$$

$$4x_1 + 5x_2 + 6x_3 = 0$$

$$7x_1 + 8x_2 + 9x_3 = 0$$

Find Null space for this

$$\begin{array}{l} r_1 \rightarrow \\ r_2 \rightarrow \\ r_3 \rightarrow \end{array} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 4 & 5 & 6 & 0 \\ 7 & 8 & 9 & 0 \end{array} \right] \underline{x} = \underline{0}$$

$\begin{array}{ccc} | & | & | \\ c_1 & c_2 & c_3 \end{array}$

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$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{bmatrix}$$

check if x is $\mathbb{R}^3$???

No.

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_3 \\ -2x_3 \\ x_3 \end{bmatrix}$$

$$= x_3 \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

$\rightarrow$ line in $\mathbb{R}^3$

$$= x_3 v$$

$$\text{span} \{ v \}$$

$$\text{col } A = \text{span} \{ \underline{c}_1, \underline{c}_2, \underline{c}_3 \}, \text{Row } A = \{ \underline{r}_1, \underline{r}_2, \underline{r}_3 \}$$

three elements

$$C = \begin{bmatrix} \begin{pmatrix} 1 \\ 5 \\ 9 \end{pmatrix} & \begin{pmatrix} 2 \\ 6 \\ 1 \end{pmatrix} & \begin{pmatrix} 3 \\ 7 \\ 2 \end{pmatrix} & \begin{pmatrix} 4 \\ 8 \\ 3 \end{pmatrix} \end{bmatrix}$$

$$\text{Null } B \in \mathbb{R}^4, \text{ col } B = \mathbb{R}^3$$

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Q Does $\begin{bmatrix} -1 \\ 2 \\ 4 \end{bmatrix}$ belong to $\text{col } A$?

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \\ 4 \end{bmatrix} = \begin{bmatrix} -1+4+12=15 \\ \vdots \\ \vdots \end{bmatrix}$$

↳ NOT in $\text{col } A$

Exp:

Find A such that $\text{col } A = W$, where $W = \text{span}\{\text{columns of } A\}$

$$W = \left\{ \begin{bmatrix} 6a-b \\ a+b \\ -7a \end{bmatrix} : a, b \in \mathbb{R} \right\}$$

check if

$$\begin{matrix} a=0, b=1 \\ \uparrow \end{matrix}$$

$$A = [\tau(e_1) \ \tau(e_2)]$$

$$\downarrow$$

$$a=1, b=0$$

$$\begin{bmatrix} 6a \\ a \\ -7a \end{bmatrix} + \begin{bmatrix} -b \\ b \\ 0 \end{bmatrix} = a \begin{bmatrix} 6 \\ 1 \\ -7 \end{bmatrix} + b \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

$$\text{span} \left\{ \begin{bmatrix} 6 \\ 1 \\ -7 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right\} = W$$

$$A = \begin{bmatrix} 6 & -1 \\ 1 & 1 \\ -7 & 0 \end{bmatrix} \xrightarrow{\text{col } A}$$

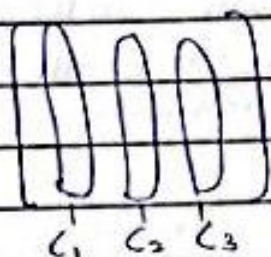
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Why study Null, row & column space?
↳ part of vector spaces

Relation diff b/w $\text{Nul } A$ & $\text{col } A$

4.



For $\text{Nul } A$ solve

$$Ax = 0$$

$$c_1 x_1 + c_2 x_2 + c_3 x_3$$

5. To find if x in $\text{Nul } A$: To solve, do $Ax = v_i$

$$Ax = 0$$

Kernel of T :

↳ All x that map to 0 in other words.

$$T(x) \rightarrow 0 = \text{Nul } A$$

$$Ax = 0$$

Range of T : Col

↳ linear combination of cols of A

$$Ax = b$$

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Ex 4.5

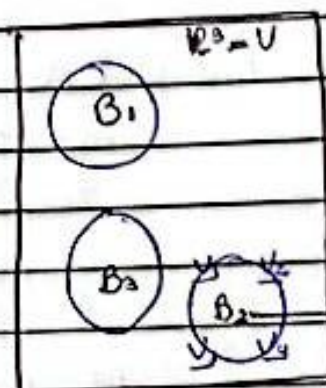
Basis / Bases

one

many

↳ A set of vectors that can cover the entire vector space.

↳ We can't use all vectors as they are infinite



Basis

1) span $\{B_1, B_2, B_3\} = V$

↳ most efficient

2) elements of B_2 are LI

↳ Some sub-spaces can do only one of the two but then they wouldn't be basis

IL:

$$v_i \neq f(v) \\ \rightarrow A \mathbf{x} = \mathbf{0}$$

Exp

Elements / Vectors of an $\mathbb{R}^3$

$$\begin{bmatrix} e_1 & e_2 & e_3 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3 \times 3}$$

↳ Basis of $\mathbb{R}^3$? ① ✓
② ✓

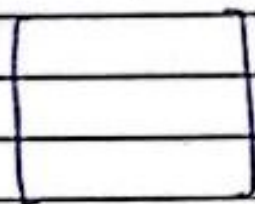
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Exp 2:

Do cols of a 3×3 invertible matrix form a basis of $\mathbb{R}^3$?

Yes

 $\hookrightarrow$ cols of $2I$ ① ✓ $\hookrightarrow$ spans $\mathbb{R}^3$ ② ✓ 3×3

Exp 3:

$$\underline{v}_1 = \begin{bmatrix} 3 \\ 0 \\ -6 \end{bmatrix}, \underline{v}_2 = \begin{bmatrix} -4 \\ 1 \\ 7 \end{bmatrix}, \underline{v}_3 = \begin{bmatrix} -4 \\ 1 \\ 7 \end{bmatrix} = \underline{v}_2$$

Do $\{\underline{v}_1, \underline{v}_2, \underline{v}_3\}$ form basis for $\mathbb{R}^3$?

Yes!

$$Ax = 0$$

$$A \sim \begin{bmatrix} \textcircled{3} & 4 & -2 \\ 0 & \textcircled{1} & 1 \\ 0 & 0 & \textcircled{2} \end{bmatrix} \quad \begin{matrix} \textcircled{1} \checkmark \\ \textcircled{2} \checkmark \end{matrix}$$

Exp 4:

$$\underline{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \underline{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \underline{v}_3 = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}$$

$\text{Span}\{\underline{v}_1, \underline{v}_2, \underline{v}_3\} = \text{Span}\{\underline{v}_1, \underline{v}_2\}$ Yes!

$$\underline{v}_3 = -\underline{v}_1 + 2\underline{v}_2$$

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Basis

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

There are following

vector spaces

1. $\text{Nul } A$, $\text{col } A$, $\text{row } A$

1) $\text{Nul } A$: $Ax = 0$

$$\begin{array}{l} \vec{r}_1 \rightarrow \\ \vec{r}_2 \rightarrow N \\ \vec{r}_3 \rightarrow \end{array} \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{bmatrix} = B \Rightarrow x = \begin{bmatrix} x_3 \\ -2x_3 \\ x_3 \end{bmatrix}$$

$$x_3 \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

Nul A

$$\text{Nul } A = \text{span} \{v\} \Rightarrow \text{Vector space of Nul } A$$

Basis of $\text{Nul } A$: $\{v\}$ → The subset of vectors space that can be used to cover vector space of $\text{Nul } A$

made with free variable

1) Row space of A :

$$\text{span} \{\vec{r}_1, \vec{r}_2, \vec{r}_3\} = \text{span} \{\vec{r}_1, \vec{r}_2\}$$

Basis of row A :

Non-zero rows of the echelon form

vectors in B by span of B can also be obtained by span of A .

GALAXY

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$$x_1 = \bar{c}_1$$

$$x_2 = x_2 - 4x_1$$

$$x_3 = c_1 x_1 + c_2 x_2 + c_3 x_3$$

Here,

linear combination of x_1, x_2 obtain

$\bar{x}_1, \bar{x}_2$

$$\text{span}\{x_1, x_2\} = \text{span}\{\bar{x}_1, \bar{x}_2\}$$

Basis of col A =

Pivot columns

form of A.

2x3:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \sim \begin{bmatrix} \boxed{1} & 0 & -1 \\ 0 & \boxed{1} & 2 \\ 0 & 0 & 0 \end{bmatrix} = B$$

$c_1 \quad c_2 \quad c_3$

$$x_1 = x_3$$

$$x_2 = -2x_3, x_1 = x_3$$

$$x_3 = x_3$$

$$\begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

$Bx = 0$ is sol of $Ax = 0$

$$Av = 0 \quad (2) \quad \quad \quad (1) \quad Bu = 0$$

holds for both A & B.

$$A = \begin{bmatrix} \underline{c}_1 & \underline{c}_2 & \underline{c}_3 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} = 0 \Rightarrow \underline{c}_1 - 2\underline{c}_2 + \underline{c}_3 = 0$$

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$$\underline{c_1 + c_3 = 2c_2}$$

$$\text{col } A \neq \text{col } B$$

∴

Row operations changed
the column space.

$$\text{col } B = \text{span} \{ \bar{c}_1, \bar{c}_2, \bar{c}_3 \} \rightarrow \text{Third value is never non-zero}$$

Basis of $\text{col } A \neq$ Basis of $\text{col } B$

↓ ↓
columns of A correspond to $\text{col } B$ Pivot cols of B
 $\{ c_1, c_2 \}$ $\{ \bar{c}_1, \bar{c}_2 \}$

Where $\{ c_1, c_3 \}$
is also correct as
they are also L.I.

* linear independ-
remains same in
 $A \neq B$.

$$\text{Nul } A = \text{Nul } B$$

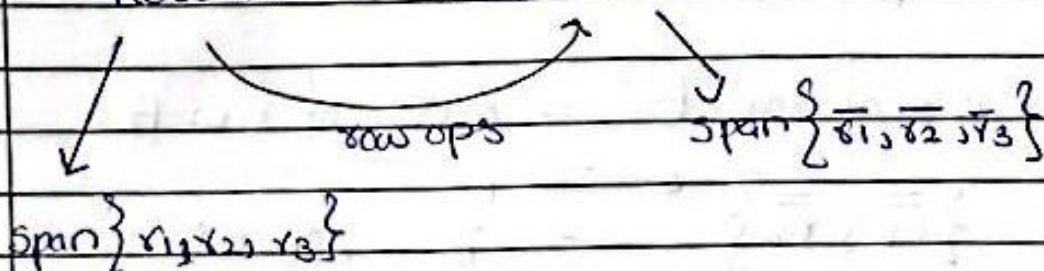
on (1) & (2).

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Row space:

Row A = Row B



$$c_1 r_1 + c_2 r_2 + c_3 r_3$$

$$\bar{c}_1 \bar{r}_1 + \bar{c}_2 \bar{r}_2 + \bar{c}_3 \bar{r}_3$$

$$\bar{c}_1 (a_{11} + a_{21} r_2 + a_{31} r_3) + \dots$$

relationship on solves
comes in the form of
 r_1, r_2, r_3

Exp

$$\begin{array}{l} r_1 \left[\begin{array}{ccc|c} 1 & 2 & 3 & r_1 \end{array} \right] \\ r_2 \left[\begin{array}{ccc|c} 4 & 5 & 6 & r_2 - 4r_1 \rightarrow r_2 \end{array} \right] \\ r_3 \left[\begin{array}{ccc|c} 7 & 8 & 9 & r_3 - 7r_1 \rightarrow r_3 \end{array} \right] \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 3 & \bar{r}_1 \\ 0 & -3 & -6 & \bar{r}_2 \\ 0 & -6 & -12 & \bar{r}_3 \end{array} \right] \quad \begin{array}{l} \bar{r}_2 - 2\bar{r}_1 \rightarrow \bar{r}_2 \\ \bar{r}_3 - 2\bar{r}_2 \rightarrow \bar{r}_3 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 3 & \bar{r}_1 = r_1 \\ 0 & -3 & -6 & \bar{r}_2 = r_2 - 4r_1 \\ 0 & 0 & 0 & \bar{r}_3 = r_3 - 2\bar{r}_2 = (r_3 - 7r_1) - 2(r_2 - 4r_1) \end{array} \right]$$

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$$\text{span} \{ \vec{v}_1, \vec{v}_2, \vec{v}_3 \} = \text{span} \{ x_1, x_2, x_3 \}$$

$$\text{Row } B = \text{Row } A$$

$$\text{Basis of row } B = \text{Basis of row } A$$

Non zero
rows of
B

$$\{ \vec{v}_1, \vec{v}_2 \} \leftarrow \{ x_1, x_2 \}$$

↳ can't find correspondence
blw x_1 & x_2 as rows can
be swapped so we take
basis of B as this.

To check row independence

$$A^T x = 0$$

Ex 4.4 Coordinate Systems

↳ representing vectors
as coordinates of basis.

$$\begin{pmatrix} v \\ \text{B} \end{pmatrix} \begin{pmatrix} \text{E} \end{pmatrix}$$

$$B = \{ b_1, b_2, b_3 \}$$

$$v = c_1 b_1 + c_2 b_2 + c_3 b_3$$

coordinates

$$\begin{bmatrix} v \\ B \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

Example,

$$E = \{ e_1, e_2 \} \rightarrow \text{standard basis}$$

$$B = \{ b_1, b_2 \}$$

$$\begin{matrix} a(1,0) \\ b(1,2) \\ e_1 \uparrow \\ e_2 \rightarrow \end{matrix}$$

$$x = \begin{bmatrix} 1 \\ 6 \end{bmatrix} = 1e_1 + 6e_2 = [e_1 \ e_2] \begin{bmatrix} 1 \\ 6 \end{bmatrix}$$

↳ coordinates of x w.r.t standard Basis $\{e_1, e_2\}$

Represent x w.r.t to basis B

$$x = c_1 b_1 + c_2 b_2$$

$$b_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, b_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$[b_1 \ b_2] \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 6 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 2 & 6 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 1 & 3 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 0 & -2 \\ 0 & 1 & 3 \end{array} \right]$$

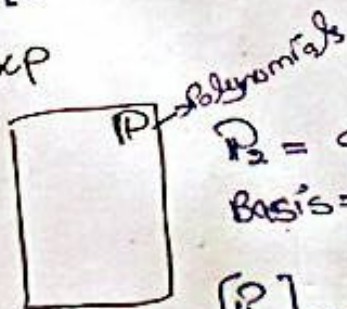
$$c_1 = -2, c_2 = 3$$

Vector space K basis is NOT unique.

$$x = P_E [x]_E = P_B [x]_B = P_C [x]_C$$

$$\begin{bmatrix} 1 \\ 6 \end{bmatrix} = [e_1 \ e_2] \begin{bmatrix} 1 \\ 6 \end{bmatrix} = [b_1 \ b_2] \begin{bmatrix} -2 \\ 3 \end{bmatrix}$$

Exp



$$P_2 = a_0 + a_1x + a_2x^2$$

Basis = $\{1, x, x^2\}$ → span can give any polynomial
→ LI

$$[P]_B = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix}$$

Use this coordinates to see whether the following polynomials are linearly independent.

$$P_1 \rightarrow 1 + 2t^2, P_2 \rightarrow 4 + t + 5t^2, P_3 \rightarrow 3 + 2t$$

$$[P_1]_B = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, [P_2]_B = \begin{bmatrix} 4 \\ 1 \\ 5 \end{bmatrix}, [P_3]_B = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 4 & 3 \\ 0 & 1 & 2 \\ 2 & 5 & 0 \end{bmatrix} \Rightarrow Ax = 0$$

NOT LI

$$x = \begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix} \Rightarrow 5c_1 + c_3 = 2c_2$$

$$\uparrow$$

$$= 5c_1 + 7c_2 + c_3 = 0$$

$$P_C x_C = P_B x_B$$

$$x_B = P_C^{-1} P_B [x]_B$$

$$[P_C^{-1} b_1, P_C^{-1} b_2] [x]_B = [x]_C$$

$$\begin{bmatrix} [b_1]_C & [b_2]_C \end{bmatrix} [x]_B = [x]_C$$

$$\downarrow$$

$$P_{C \leftarrow B}$$

$$\begin{bmatrix} [c_1]_B & [c_2]_B \end{bmatrix} [x]_C = [x]_B$$

$$x = P_{E \leftarrow C} [x]_C$$

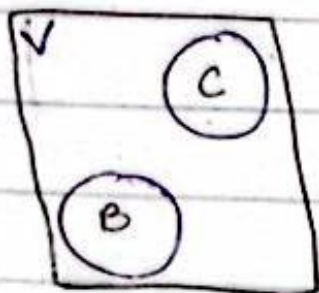
$$P_C = P_{E \leftarrow C}$$

$$[c_1, c_2] = [P_C^{-1} c_1, P_C^{-1} c_2]$$

$$\downarrow$$

$$P_E = P_C^{-1}$$

Ex 4.6

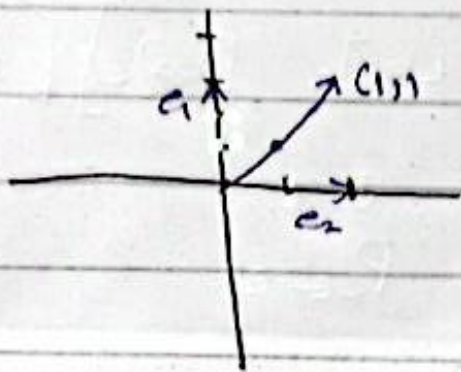


$[0], [1]$

$$B = \{\underline{b}_1, \underline{b}_2\}, C = \{\underline{c}_1, \underline{c}_2\} \rightarrow [0], [1]$$

$$E = \{\underline{e}_1, \underline{e}_2\}$$

$$\underline{x} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$$



$$= 1\underline{e}_1 + 4\underline{e}_2$$

$$= x^B \underline{b}_1 + x_2^B \underline{b}_2$$

$$= \underbrace{\begin{bmatrix} \underline{b}_1 & \underline{b}_2 \end{bmatrix}}_{P_B} \begin{bmatrix} x_1^B \\ x_2^B \end{bmatrix}$$

$$P_B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$x_B = P_B^{-1} \underline{x} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$$

$$P_B^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

$$\underline{x} = -3\underline{b}_1 + 4\underline{b}_2$$

$$P_E \underline{x}_E - P_B [\underline{x}]_B = P_C [\underline{x}]_C$$

$$\underline{x} =$$

Day:

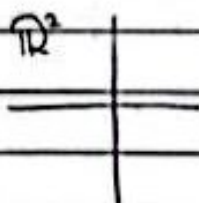
Math contin

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Ex 4.5

Dimension of a Vector Space

Intrinsic property $\Rightarrow$ like 'Blood Group' of a vector space for Humans



$$\dim \mathbb{R}^n = n$$

$$\dim \mathbb{P}_n = n+1$$

$$\mathbb{P}_2 = a_0 + ax + ax^2$$

$$\dim \{V = \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 6 \end{bmatrix} \right\} \} = 2$$

linearly dependent.

Only one can be LI ~~the other~~ $\therefore$ thus only a line.

Dimension of a vector space = no. of vectors in its Basis.

(Same for all basis)

Day:

no. of vectors in $\text{Col } A$ no. of vectors in $\text{Row } A$

Basis of $\text{Col } A = \text{Basis of Row } A$

no. of pivot cols of $\text{Col } A$ no. of non-zero of A echelon

$\text{Dim of Col } A = \text{Dim of Row } A = \text{Rank } A$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{bmatrix} \begin{matrix} r_1 \\ r_2 \\ r_3 \end{matrix}$$

$c_1 \quad c_2 \quad c_3$

Basis of $\text{Col } A = \{c_1, c_2\}$ Basis of $\text{Row } A = \{r_1, r_2\}$

$\text{Dim Nul } A = \text{Nullity}$

$\text{Rank} + \text{Nullity} = \# \text{ of cols of } A$

Exp

$\text{Rank } A = 2, \text{ Nullity} = 1$

$2 + 1 = 3 = \# \text{ of cols of } A$

$\text{Nullity} = 2 \quad [A]_{9 \times 9}$

$\text{Rank} = 7$

Could a 6×9 matrix have nullity 2?

No!

$\text{max Rank} = 6$

$2 + 7 = 9$

Col Rank

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Example

A scientist...

$$Ax = 0$$

$$[A]_{40 \times 42}$$

2 solution

$$\text{Nullity} = 2$$

$$\text{Rank} = 40$$

Thus, each row has a pivot. So,

$Ax = b$ has a solution

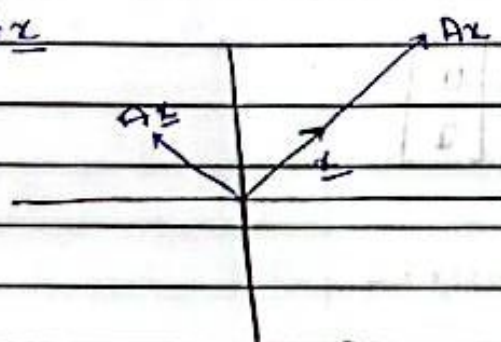
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Chapter 05 Eigenvalues & Eigenvectors

↳ German
↳ "Characteristics"

$$b = Ax$$



$$Ax = \lambda x$$

↳ eigenvalue
↳ eigenvectors, non-zero

Some x on which A acts to just scale it & direction remains same.

What are the eigen-vectors of I_n ?

↳ All vectors except 0

↳ Same for $\begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}$

$$\begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} kx \\ ky \end{bmatrix}$$

$$= k \begin{bmatrix} x \\ y \end{bmatrix}$$

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↳ Eigen-space is all eigenvectors for λ

$$\begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix}$$

, $\lambda = 2$ is eigenvalue

For λ :

$$(A - \lambda I)x = 0$$

$$\begin{bmatrix} 2 & -1 & 6 & | & 0 \\ 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

↳ How many LI eigenvectors in the solution of above tells dimension of eigen-space

$$x_2 \begin{bmatrix} 1/2 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}$$

✓ collection of all eigenvector

Span is eigen vector

↳ Eigen-values of a triangular matrix are its entries on the diagonal.

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

λ_1
 λ_2

$$Ax = \lambda x$$

For $\lambda = 0$

For this, $\lambda = 0$ then A is NOT invertible

$$Ax = 0$$

$$x = A^{-1} \cdot 0$$

$= 0$ vector can't be eigenvector.

← If it is we will have x as 0 but

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$x_0 = \text{initial population}$

$$x_{k+1} = Ax_k$$

$$x_{500} = A^{500} x_0$$

but if

$x_0 = \text{eigen-vector}$

then

$$x_1 = Ax_0 = \lambda x_0$$

$$x_2 = Ax_1$$

$$= A(\lambda x_0)$$

$$= \lambda Ax_0$$

$$= \lambda \lambda x_0$$

$$= \lambda^2 x_0$$

Day: Ex 5.2: Characteristic Eq Date: / /

Where/Why we use?

↳ To find eigen-values

$$(A - \lambda I)x = 0$$

if $(A - \lambda I) \neq 0$

then x is $\vec{0}$ which is NOT possible for eigen-vectors.

So,

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 2-\lambda & 3 \\ 3 & -6-\lambda \end{vmatrix} = 0$$

$$(2-\lambda)(-6-\lambda) - 9 = 0$$

$$\lambda^2 + 4\lambda - 22 = 0$$

$$\lambda_1 = 3, \lambda_2 = -7$$

$$\lambda = \sqrt{0}$$
$$\lambda = 0, 0, 0$$

Exp

$$\lambda^6 (-4\lambda^5 - 12\lambda^4) = 0$$

A row ops echelon A

$$\lambda^4 (\lambda^2 - 4\lambda - 12) = 0$$

↳ row ops don't preserve eigen-values

↓
0 with multiplicity 4

↓
How many times 0 comes

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Step

$$-\lambda^3 - 3\lambda^2 + 4 = 0$$

Trail solution

$$\lambda = 1$$

→ solve

$$\lambda - 1 \mid -\lambda^3 - 3\lambda^2 + 4$$

⋮

$$\overbrace{PAP^{-1}}^{\text{similarity transformation}} = B$$

Then, A & B are similar

↳ Preserve some properties

↳ Rank

↳ Eigen-values

↳ Trace

↳ Determinant

$$\text{Diagonalization: } PAP^{-1} = D$$

$$A = PBP^{-1}$$

→ change operation in simplified manner

$$Ax = PBD^{-1}x \quad \text{change of basis}$$

coming back to original basis
going to new basis

$$P^{-1}Ax = \lambda x$$

GALAXY

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Ex 5.3 Diagonalization

$$A = P D P^{-1}$$

↳ Diagonal Matrix

$$A^2 = A \cdot A$$

$$= (P D P^{-1}) (P D P^{-1})$$

$$= P D^2 P^{-1}$$

$$D = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$$

$$P = [\text{eigenvector}_1 \dots \text{eigenvector}_n]$$

↳ columns are LI cuz P are Invertible

$$AP = PD$$

$$A [\underline{v}_1 \ \underline{v}_2 \ \dots \ \underline{v}_n] = [A \underline{v}_1 \ A \underline{v}_2 \ \dots \ A \underline{v}_n]$$

$$= [\lambda_1 \underline{v}_1 \ \lambda_2 \underline{v}_2 \ \dots \ \lambda_n \underline{v}_n]$$

$$= [\underline{v}_1 \ \underline{v}_2 \ \dots \ \underline{v}_n] \begin{bmatrix} \lambda_1 & 0 \\ & \ddots \\ 0 & \lambda_n \end{bmatrix}$$

$$P \quad D$$

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Exp

$$A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$$

$$\det(A - \lambda I) = 0$$

$$\det \begin{bmatrix} 1-\lambda & 2 \\ 4 & 3-\lambda \end{bmatrix} = 0$$

$$\lambda^2 - 4\lambda - 5 = 0$$

$$\lambda_1 = -1, \lambda_2 = 5$$

For $\lambda_1 = -1$;

$$\begin{bmatrix} 2 & 2 \\ 4 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} 2 & 2 & 0 \\ 4 & 4 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$x_1 = -x_2$$

$$x = \begin{bmatrix} -x_2 \\ x_2 \end{bmatrix}$$

$$= x_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

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For $\lambda_2 = 5$;

$$\begin{bmatrix} -4 & 2 & 0 \\ 4 & -2 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} -2 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$x = x_2 \begin{bmatrix} 1/2 \\ 1 \end{bmatrix}$$

$$= x_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

As $n=2$ & we have 2 LI vectors, so we can proceed with Diagonalization

$$D = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ 0 & -1 \end{bmatrix}$$

$$P = \begin{bmatrix} \underline{v_2} & \underline{v_1} \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}$$

Exp

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

$\lambda_1 = 1$ with multiplicity 2

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For $\lambda = 1$,

$$\begin{bmatrix} 0 & 1 & : & 0 \\ 0 & 0 & : & 0 \end{bmatrix}$$

$$\underline{x} = x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Exp

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$\lambda_1 = 1$ with multiplicity 2

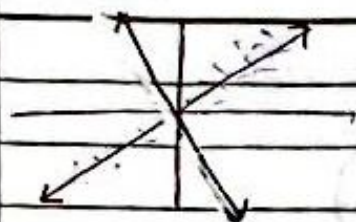
$$\begin{bmatrix} 0 & 0 & : & 0 \\ 0 & 0 & : & 0 \end{bmatrix}$$

$$\underline{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

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PCA changes the bases from e to e ^{bases of} eigen-vector
Damn!!!

Ex 5.4 Eigenvectors & Linear Transformation

$$T(ax_1 + bx_2) = aT(x_1) + bT(x_2)$$

$$T(x) = Ax$$

$$T(x) = \lambda x$$

eigen-values &

such vectors are eigen-vectors of
this linear transformation

$$x \xrightarrow{T} T(x)$$



$$[x]_B$$

which?

$$[T(x)]_B$$

$$\text{Basis} = B = \{b_1, b_2, \dots, b_n\}$$

$$x = x_1^B b_1 + x_2^B b_2 + \dots + x_n^B b_n \quad \text{①}$$

$$[x]^B = \begin{bmatrix} x_1^B \\ \vdots \\ x_n^B \end{bmatrix}$$

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Using (1):

$$T(x) = T(x_1^B b_1 + \dots + x_n^B b_n)$$

$$= x_1^B T(b_1) + \dots + x_n^B T(b_n)$$

$$= [T(b_1) \ T(b_2) \ \dots \ T(b_n)] \begin{bmatrix} x_1^B \\ \vdots \\ x_n^B \end{bmatrix}$$

$$[T(x)]_B = \begin{bmatrix} [T(b_1)]_B & [T(b_2)]_B & \dots & [T(b_n)]_B \end{bmatrix} \begin{bmatrix} x_1^B \\ \vdots \\ x_n^B \end{bmatrix}$$

$$M = [T]_B$$

$$[T(x)]_B = [T]_B [x]_B$$

$\therefore B$ - matrix of the transformation T

Ex

If $B = \{b_1, b_2\}$ is a basis for V

Let $T: V \rightarrow V$ be the linear transformation such that

$$T(b_1) = 3b_1 - 2b_2, \quad T(b_2) = 4b_1 + 7b_2$$

Find the matrix M for this transformation T relative to the basis B

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$$M = \begin{bmatrix} [T(b_1)]_B & [T(b_2)]_B \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 4 \\ -2 & 7 \end{bmatrix}$$

Exp 2:

Let $T: \mathbb{P} \rightarrow \mathbb{P}_2$ given by

$$T(a_0 + a_1 t + a_2 t^2) = a_1 + 2a_2 t \quad \text{be a LT.}$$

(a) Find B-Matrix for $T \Rightarrow M = [T]_B = \begin{bmatrix} [T(b_1)]_B & [T(b_2)]_B \end{bmatrix}$ (b) Verify $[T(p)]_B = M[p]_B$

$$B = \{1, t, t^2\}$$

$$T(b_1) = 0, \quad T(b_2) = 1, \quad T(b_3) = 2t$$

$$[T(b_1)]_B = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$[T(b_2)]_B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$[T(b_3)]_B = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}$$

$$M = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

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$$p = a_0 + a_1 t + a_2 t^2$$

$$[p]_B = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix}$$

$$\text{LHS} = T(p) = a_1 + 2a_2 t$$

$$\text{RHS} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix}$$

$$[T(p)]_B = \begin{bmatrix} a_1 \\ 2a_2 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} a_1 \\ 2a_2 \\ 0 \end{bmatrix}$$

$$\text{LHS} = \text{RHS}$$

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Chapter #6

Ex 6.1

$$Ax = b$$

↳ Solution to inconsistent system (Approximate it)

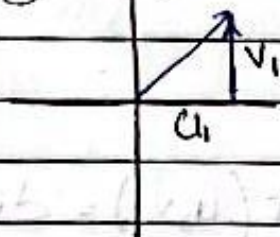
A. Inner Product: (Dot Product)

$$u \cdot v = u_1v_1 + u_2v_2 + u_3v_3 = \sum_{i=1}^n u_i v_i$$

$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \cdot \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = u^T v = \text{Scalar (Ans)}$$

$$u \cdot v = v \cdot u$$

Length of a vector



$$u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\|u\| = \sqrt{u_1^2 + u_2^2}$$

$$u \cdot u = u_1^2 + u_2^2$$

$$\|u\| = \sqrt{u \cdot u}$$

$$\hat{u} = \frac{u}{\|u\|}$$

$$\|\hat{u}\| = \frac{1}{\|u\|} \|u\| = 1$$

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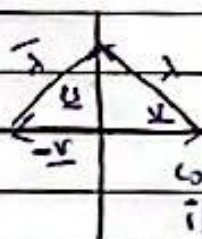
Distance in $\mathbb{R}^2$



$$\text{dist}(u, v) = \|u - v\|$$

$$= \|v - u\|$$

Orthogonality: $\perp$
 (perpendicular)



trying to tell
 if $\perp$ but in $\mathbb{R}^n$
 it is difficult

$u \perp v$ if and only if $\text{dist}(u, v) = \text{dist}(u, -v)$

$$\|u - v\| = \|u - (-v)\|$$

$$\|u - v\| = \|u + v\|$$

$$(u - v) \cdot (u - v) = (u + v) \cdot (u + v)$$

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$$\cancel{u \cdot u} + \cancel{v \cdot v} + 2\cancel{u \cdot v} = \cancel{u \cdot u} + \cancel{v \cdot v} + 2\cancel{u \cdot v}$$

$$-u \cdot v = u \cdot v$$

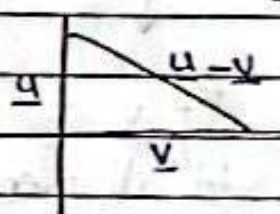


Possible iff $u \cdot v = 0$

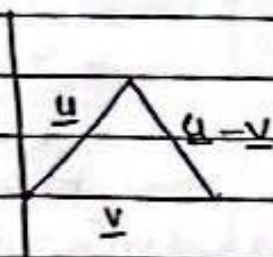


orthogonality

Pythagorean theorem:
 For right-angle triangle

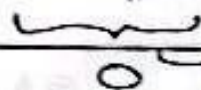


$$\|u-v\|^2 = \|u\|^2 + \|v\|^2$$



Laws of cosines (Kashi's theorem)

$$\|u-v\|^2 = \|u\|^2 + \|v\|^2 - 2\|u\|\|v\|\cos\theta$$



$\theta = 90^\circ$

$\cos 90^\circ = 0$ GALAXY

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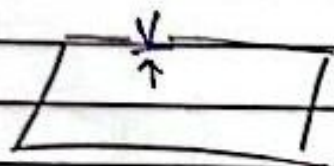
$$(u-v) \cdot (u-v)$$

$$u \cdot u + v \cdot v - 2u \cdot v$$

$$\|u\|^2 + \|v\|^2 - 2u \cdot v = \|u\|^2 + \|v\|^2 - 2\|u\|\|v\|\cos\theta$$

$$u \cdot v = \|u\|\|v\|\cos\theta$$

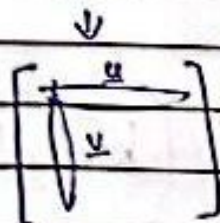
Vector orthogonal to a sub-space W of $\mathbb{R}^n$



If v is orthogonal to all elements of W

$W^\perp =$ Set of All vectors orthogonal to W
 $\downarrow$
 W -complement

$$(\text{Row } A)^\perp = \text{col } A$$



$$u \cdot v \neq 0$$

$$= \text{Nul } A \checkmark$$

$$Ax = 0$$

$$\begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} x = 0$$

$$v_1 x = 0$$

$$v_2 x = 0$$

$$v_n x = 0$$

$$\text{Nul } A = (\text{Row } A)^\perp$$

GALAXY

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$$\begin{bmatrix} \\ \\ \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x_1^T \cdot x^w = 0$$

$$x_2^T \cdot x^w = 0$$

$$(\text{Col } A)^\perp = (\text{Nul } A')$$

Ex 6.2

$$\{v_1, v_2, \dots, v_n\}$$

Orthogonal set (OS)

$$\text{iff } v_i \cdot v_j = 0 \text{ if } i \neq j$$

Exp

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, v_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, B = \{v_1, v_2, v_3\} \rightarrow \text{OS}$$

Exp

$$b_1 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, b_2 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$B = \{b_1, b_2\} \Rightarrow \text{OS}$$

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Orthogonality $\Rightarrow$ Linear Independence

$\leftarrow$
not true

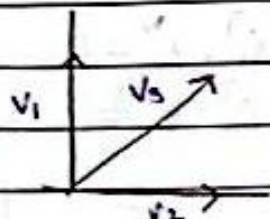
$$B = \{b_1, b_2, \dots, b_n\}$$

$\hookrightarrow$ Bases

$\hookrightarrow$ Orthogonal

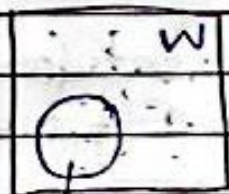
then

Orthogonal Basis



$$v_1 \perp v_2 \Rightarrow \text{O.S.} \Rightarrow \text{L.I.}$$

$$v_3 \not\perp v_2 \Rightarrow \text{L.I.}$$



ortho-basis
(e)

$$y = c_1 b_1 + c_2 b_2 + \dots + c_n b_n \quad \text{--- ①}$$

What benefit of ortho-basis?

before find $c_1, c_2, \dots, c_n$ by

$$y = [b_1 \dots b_n] \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$$

but now,

$$c_i = \frac{y \cdot b_i}{b_i \cdot b_i}$$

prod

dot / both sides ① by b_1 ;

$$y \cdot b_1 = c_1 \underbrace{b_1 \cdot b_1}_0 + c_2 \underbrace{b_2 \cdot b_1}_0 + \dots + c_n \underbrace{b_n \cdot b_1}_0$$

$$c_1 = \frac{y \cdot b_1}{b_1 \cdot b_1}$$

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Exp:

If M is an orthogonal basis with $M = \{ \underline{v}_1, \underline{v}_2, \underline{v}_3 \}$ when

$$\underline{v}_1 = \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix}, \underline{v}_2 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, \underline{v}_3 = \begin{bmatrix} -1/2 \\ -2 \\ 7/2 \end{bmatrix}$$

Express $y = \begin{bmatrix} 6 \\ 1 \\ -8 \end{bmatrix}$ as linear combination of vectors in V

$$y = c_1 \underline{v}_1 + c_2 \underline{v}_2 + c_3 \underline{v}_3$$

$$c_1 = \frac{y \cdot \underline{v}_1}{\underline{v}_1 \cdot \underline{v}_1}$$

$$= \frac{6(3) + 1(1) + (-8)(-1)}{3(3) + 1(1) + (-1)(-1)} = \frac{11}{11} = 1$$

$$c_2 = \frac{y \cdot \underline{v}_2}{\underline{v}_2 \cdot \underline{v}_2} = -2, \quad c_3 = \frac{y \cdot \underline{v}_3}{\underline{v}_3 \cdot \underline{v}_3} = 1$$

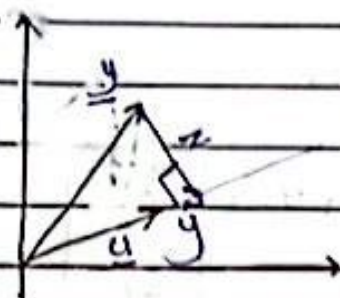
$$\underline{y} = \underline{u}_1 - 2\underline{u}_2 - 2\underline{u}_3$$

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Orthogonal Projection

split into 2
 - one in sub-space
 - one $\perp$ to sub-space
 $y \in$ a sub-space
 (line) $W = \text{span}\{u\}$



Project y onto u .

$$y = \hat{y} + z$$

$$= \alpha u + z \Rightarrow \hat{y} \perp z$$

$$y \cdot u = \alpha u \cdot u + z \cdot \underline{\hat{u}}^{\perp}$$

$$\alpha = \frac{y \cdot u}{u \cdot u}$$

unique

Exp:

$$\text{Let } y = \begin{bmatrix} 7 \\ 6 \end{bmatrix} \text{ and } u = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

Write y as the sum of 2 orthogonal vectors, one is in $\text{span}\{u\} = W$ & other orthogonal to u

$$y = \alpha u + z$$

$$\alpha = \frac{y \cdot u}{u \cdot u}$$

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$$= \frac{7(4) + 6(2)}{4(4) + 2(2)}$$

$$= 2$$

$$\underline{\underline{y}} = \underline{\underline{2u}} + \underline{\underline{z}}$$

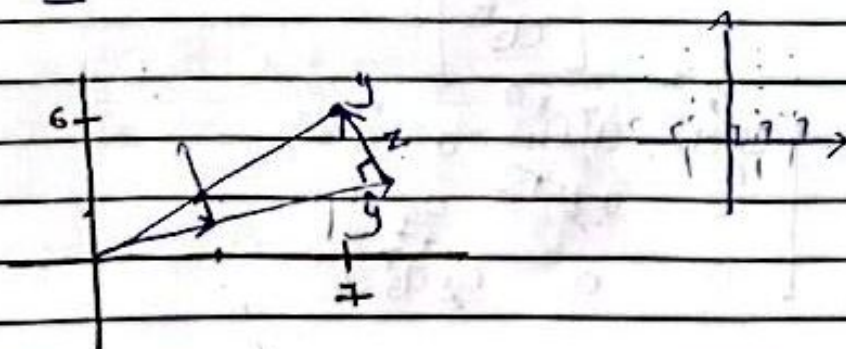
$$\underline{\underline{z}} = \underline{\underline{y}} - \underline{\underline{y}}$$

$$= \begin{bmatrix} 7 \\ 6 \end{bmatrix} - \begin{bmatrix} 8 \\ 4 \end{bmatrix}$$

$$= \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$$\underline{\underline{z}} \cdot \underline{\underline{\hat{y}}} = \underline{\underline{z}} \cdot \underline{\underline{u}} = 0$$

$$\begin{bmatrix} -1 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$



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Orthonormal set vs Orthogonal set

$$\{u_1, u_2, \dots, u_n\}$$

$$\{b_1, b_2, \dots, b_n\}$$

1) $u_i \cdot u_j = 0$ for $i \neq j$

1) $b_i \cdot b_j = 0$ for $i \neq j$

2) $u_i \cdot u_i = 1$

Theorem 06:

$$[U]_{m \times n} = \begin{bmatrix} | & | & | \\ | & | & | \\ | & | & | \end{bmatrix}$$

$m \times n$ is no. of vectors in $\mathbb{R}^m$

$\underbrace{U^T \cdot U}_{\text{Inner Product}} x = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = x$
 $x \in \mathbb{R}^n$

$$[u_1 \ u_2 \ u_3] \begin{bmatrix} u_1^T \\ u_2^T \\ u_3^T \end{bmatrix}$$

$$\begin{bmatrix} \cancel{u_1^T u_1}^1 & \cancel{u_1^T u_2}^0 & 0 \\ 0 & \cancel{u_2^T u_2}^1 & 0 \\ 0 & 0 & \cancel{u_3^T u_3}^1 \end{bmatrix}$$

$U [U^T x] c \Rightarrow x \in \mathbb{R}^n$

$$Uc = [u_1 \ u_2 \dots \ u_n] \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

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$$= c_1 u_1 + c_2 u_2 + \dots + c_n u_n$$

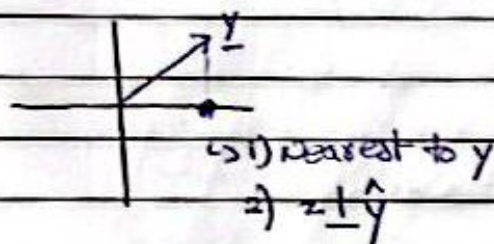
$$c = U^T x$$

$$\Rightarrow U U^T x \Rightarrow \text{proj}_U x$$

$$\Rightarrow U^T U x \Rightarrow x$$

$$\begin{bmatrix} u_1^T \\ u_2^T \\ \vdots \\ u_n^T \end{bmatrix} \begin{bmatrix} x \\ \vdots \end{bmatrix} = \begin{bmatrix} u_1^T x \\ u_2^T x \\ \vdots \\ u_n^T x \end{bmatrix}$$

$$= \begin{bmatrix} u_1 \cdot x \\ u_2 \cdot x \\ \vdots \\ u_n \cdot x \end{bmatrix}$$



Ex 6.3: Orthogonal Projections

y in $\mathbb{R}^n$ let $n=5$
Let W be sub-space of $\mathbb{R}^5$

Orthogonal Basis $\{u_1, u_2, u_3, u_4, u_5\}$

let $w \in \mathbb{R}^4$

$$y = (c_1 u_1 + c_2 u_2 + c_3 u_3 + c_4 u_4) + c_5 u_5$$

$$y = \text{proj}_W y + z, \quad z = y - \hat{y} = c_5 u_5$$

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Exp

$$\text{Let } u_1 = \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix}, u_2 = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}, \underline{y} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

If $W = \text{span} \{u_1, u_2\}$ Write $\underline{y}$ as sum of a vector in W & a vector orthogonal to W ?

$$u_1 \cdot u_2 = -4 + 5 + 1 \\ = 0$$

$$\hat{y} = \text{proj}_W y = c_1 u_1 + c_2 u_2$$

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} \quad c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} \\ = \frac{9}{30} \quad = \frac{3}{6}$$

$$\hat{y} = \text{proj}_W y = \frac{9}{30} \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix} + \frac{3}{6} \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$$

$$z = y - \hat{y}$$

$$= \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} - \hat{y}$$

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$$[u u^T] y = \begin{bmatrix} u_1 & \dots & u_n \end{bmatrix} \begin{bmatrix} u_1^T \\ u_2^T \\ \vdots \\ u_n^T \end{bmatrix}$$

$$[u] [u^T y] = [u] \begin{bmatrix} u_1^T \\ u_2^T \\ \vdots \\ u_n^T \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

$$= \begin{bmatrix} u_1 & \dots & u_n \end{bmatrix} \begin{bmatrix} y \cdot u_1 \\ \vdots \\ y \cdot u_n \end{bmatrix} \begin{matrix} \rightarrow c_1 \\ \vdots \\ c_n \end{matrix}$$

$$\text{proj}_u y = c_1 u_1 + c_2 u_2 + \dots + c_n u_n$$

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1}$$

Exp

Pattern Recognition

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \rightarrow u = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

1. Vectorize your image x
2. Calc $x^T M x = c$
3. If $c = 0$.

x is desired pattern ($x = u$)

else

Not

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Steps to calculate

A) Define a sub-space $W = \text{span} \{u\}$

B) Define " " $W^\perp = \text{orthogonal to } W$

A $\{v_1, v_2, \dots, v_p\}$ basis vectors of orthogonal $W^\perp$.

$$C) V = \begin{bmatrix} v_1^T \\ v_2^T \\ \vdots \\ v_p^T \end{bmatrix}$$

D) $\forall x = c \Rightarrow c = 0$ iff $x = u$
 $\neq 0$ else NOT

$$\|c\|^2 = c \cdot c = [v_x] [v_x]$$

$$= [v_x]^T v_x$$

$$= x^T \underbrace{[V^T V]}_M x$$

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$$\text{Exp } \underline{u} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, W = \text{span} \{ \underline{u} \}$$

$$\underline{u} \cdot \underline{x} = 0$$

$$\begin{bmatrix} 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = 0$$

$$x_1 + x_4 = 0$$

$$x_1 = -x_4$$

$$\underline{x} = \begin{bmatrix} -x_4 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

$$= x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$\underline{v}_1 \quad \underline{v}_2 \quad \underline{v}_3$

$$V = \begin{bmatrix} \underline{v}_1^T \\ \underline{v}_2^T \\ \underline{v}_3^T \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}_{3 \times 4}$$

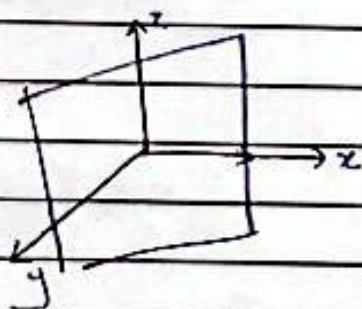
$$V^T V = \begin{bmatrix} 0 & 0 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{4 \times 3} \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}_{3 \times 4}$$

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$$= \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}$$

$$x = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

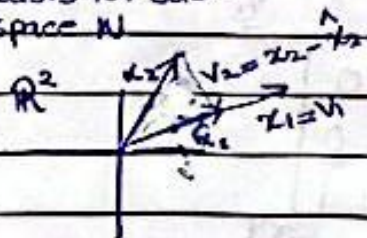


z-y plane $\perp$ x

so gives 0

Ex 6.4 The Gram-Schmidt Process

In	Algo	Out
Non-orthogonal basis for subspace W		Orthogonal basis



$$v_1 = x_1$$

$$v_2 = x_2 - \text{proj}_{x_1} x_2$$

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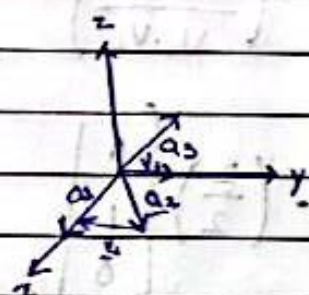
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Exp.

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

 $\underline{a}_1 \quad \underline{a}_2 \quad \underline{a}_3$

$$W = \text{span} \{ \underline{a}_1, \underline{a}_2, \underline{a}_3 \}$$



$$v_1 = a_1 \Rightarrow W_1 = \text{span} \{ v_1 \}$$

$$\underline{v}_2 = \underline{a}_2 - \text{proj}_{W_1} \underline{a}_2$$

$$= \underline{a}_2 - \text{proj}_{v_1} \underline{a}_2$$

$$= \underline{a}_2 - \alpha \underline{a}_1$$

$$= \underline{a}_2 - \left(\frac{\underline{a}_2 \cdot \underline{a}_1}{\underline{a}_1 \cdot \underline{a}_1} \right) \underline{a}_1$$

$$= \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} - \frac{1}{1} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

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$$\underline{v}_3 = \underline{a}_3 - \left(\text{proj}_{\underline{w}_2} \underline{a}_3 \right) \quad \underline{w}_2 = \text{span}\{v_1, v_2\}$$

$$= \underline{a}_3 - \left(\text{proj}_{\underline{v}_1} \underline{a}_3 + \text{proj}_{\underline{v}_2} \underline{a}_3 \right)$$

$$= \underline{a}_3 - \left(\frac{\underline{a}_3 \cdot \underline{v}_1}{\underline{v}_1 \cdot \underline{v}_1} \right) \underline{a}_3 - \left(\frac{\underline{a}_3 \cdot \underline{v}_2}{\underline{v}_2 \cdot \underline{v}_2} \right) \underline{a}_3$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \left(\frac{1}{1} \right) \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - \left(\frac{2}{2} \right) \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

If $\{a_1, a_2, \dots, a_p\}$ non orthogonal basis of $\text{Col } A$

$\{v_1, v_2, \dots, v_p\}$ orthogonal basis of $\text{Col } A$

$$A = [a_1 \ a_2 \ \dots \ a_p]$$

$\left. \begin{array}{l} a_1 = Q_1 e_1 \\ a_2 = Q_2 e_2 \end{array} \right\} \text{change of basis}$

$$Q = [v_1 \ v_2 \ \dots \ v_p]$$

$$A = [Q_1 \ Q_2 \ \dots \ Q_p]$$

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$$A = Q [r_1 \ r_2 \ \dots \ r_p]$$

$$= QR$$

↳ upper triangular

$$a_1 = Q r_1$$

$$\begin{bmatrix} v_1 & v_2 & \dots & v_p \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \rightarrow \text{To find orthogonal you will need only prev } v_1$$

$$a_2 = Q r_2$$

$$= \begin{bmatrix} v_1 & v_2 & \dots & v_p \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \rightarrow \text{need you / prev 2}$$

Gram-Schmidt process
move i -th terms to left
to get original vector
as $a_i = Q r_i$

$$A = QR$$

$$Q^T A = Q^T Q R$$

I for orthonormal

$$Q^T A = R$$

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$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Find QR factorization of A.

1. Apply Gram-Schmidt:

$$v_1 = a_1 \rightarrow W_1 = \text{span}\{a_1\}$$

$$v_2 = a_2 - \text{proj}_{W_1} a_2 \Rightarrow v_2 = \begin{bmatrix} 3/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix}$$

$$v_3 = a_3 - \text{proj}_{W_2} a_3$$

$$v'_2 = \begin{bmatrix} 3 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

 $\hookrightarrow W_2 = \text{span}\{v_1, v'_2\}$

$$v'_3 = \begin{bmatrix} 0 \\ -2/3 \\ 1/3 \\ 1/3 \end{bmatrix} \quad v'_3 = \begin{bmatrix} 0 \\ -2 \\ 1 \\ 1 \end{bmatrix}$$

To orthonormal:

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \Rightarrow q_1 = \frac{1}{\sqrt{4}} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{4} \\ 1/\sqrt{4} \\ 1/\sqrt{4} \\ 1/\sqrt{4} \end{bmatrix}$$

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$$\underline{q}_1 = \frac{1}{\|v_2\|} \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix}$$

$$\underline{q}_2 = \frac{1}{\sqrt{6}} \begin{bmatrix} 0 \\ -2 \\ 1 \\ 1 \end{bmatrix}$$

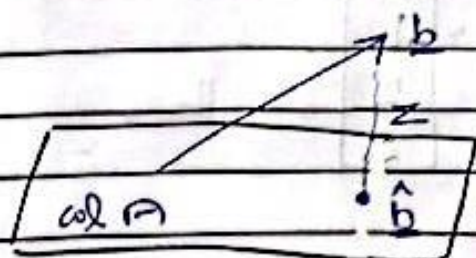
$$= \frac{1}{\sqrt{12}} \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix}$$

Ex 6.5

Problem $\rightarrow$ Inconsistent systemsLeast Square $\rightarrow$ Approx solution to above

Minimum sq of errors

$$\begin{array}{cc} Ax = b \\ \downarrow \quad \downarrow \\ \text{known} \quad \text{known} \end{array}$$



$$A\hat{x} = \hat{b}$$

so can't get b using linear combination of cols of A

proj b
col A

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$$\text{col } A \perp z$$

$$b - \hat{b}$$

$$\text{col } A \perp (b - A\hat{x})$$

columns of $A \perp (b - A\hat{x})$

$$A^T(b - A\hat{x}) = 0$$

$$A^T b = A^T A \hat{x}$$

$$\hat{x} = (A^T A)^{-1} A^T b \quad \text{if } (A^T A)^{-1} \text{ invertible}$$

↳ Unique sol

else,

$$\hat{b} = B\hat{x}$$

↳ Infinitely many solutions

$$[A]_{m \times n} [x]_{n \times 1} = [b]_{m \times 1}$$

b has same len as columns of A .

Exp.

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 2 \\ 0 & 0 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}$$

↳ Inconsistent

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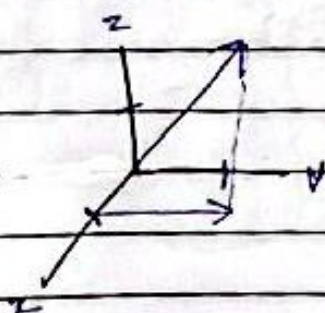
$$A^T A$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 2 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 5 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix} = b'$$

$$(A^T A) \hat{z} = \begin{bmatrix} 1 & 5 & -1 \\ 4 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$



$$A \hat{z} = \hat{b}$$

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\text{Error} = \| \hat{b} - b \|$$

$\hat{z}$ will not satisfy original system but it will to an extent with error!

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$$\text{proj}_W b = c_1 u_1 + c_2 u_2$$

$$= \left(\frac{b \cdot u_1}{u_1 \cdot u_1} \right) u_1 + \left(\frac{b \cdot u_2}{u_2 \cdot u_2} \right) u_2$$

$$\hat{x}_{\text{ortho}} = A^T b \rightarrow \text{If orthonormal then } A^T A = I$$

$$\begin{bmatrix} a_1 \cdot b \\ \vdots \\ a_n \cdot b \end{bmatrix}$$

$$\begin{aligned} QR \hat{x} &= \overbrace{Q Q^T}^I b \\ A \hat{x} &= \hat{b} \end{aligned}$$

mul by Q^T

$$\overbrace{Q^T Q}^I R \hat{x} = \overbrace{Q^T Q}^I Q^T b$$

$$R \hat{x} = Q^T b$$

$$\hat{x} = R^{-1} Q^T b$$

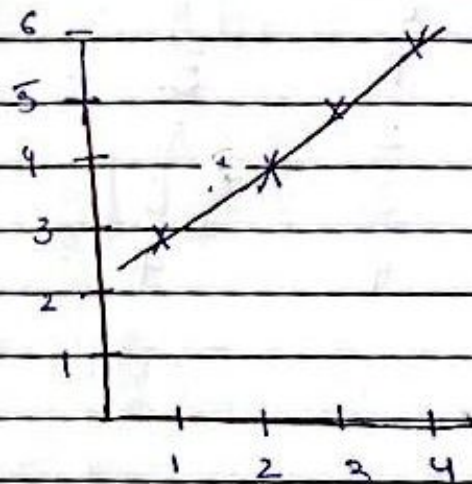
→ Numerical Methods

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Ex 6.6

x	y
1	3
2	4
3	5
4	6



$$y = mx + c$$

$$\begin{bmatrix} 3 \\ 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} x & 1 \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix}$$

$$\begin{bmatrix} 3 \\ 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix}$$

$$\begin{bmatrix} 3 & 1 & 3 \\ 4 & 1 & 4 \\ 5 & 1 & 5 \\ 6 & 1 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 3 \\ 0 & -1 & -2 \\ 0 & -2 & -4 \\ 0 & -3 & -6 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$x_2 = 2, x_1 + x_2 = 3$$

$$x_1 = 1$$

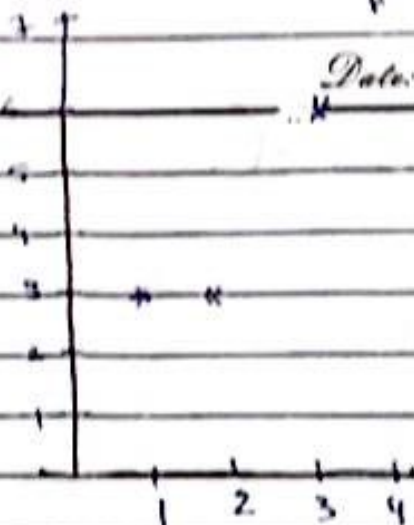
$$m = x_1 = 1$$

$$c = x_2 = 2$$

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x	y
1	3
2	3
3	6
4	7



$$\begin{bmatrix} 3 \\ 3 \\ 6 \\ 7 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix}$$

$$y = Ax$$

$$\hat{y} = A\hat{x}$$

$$(A^T A) \hat{x} = A^T y$$

$$A^T A = \begin{bmatrix} 29 & 10 \\ 10 & 4 \end{bmatrix}$$

$$(A^T A)^{-1} = \frac{1}{16} \begin{bmatrix} 4 & -10 \\ -10 & 29 \end{bmatrix}$$

$$A^T y = \begin{bmatrix} 57 \\ 19 \end{bmatrix}$$

$$\hat{x} = \frac{1}{16} \begin{bmatrix} 4 & -10 \\ -10 & 29 \end{bmatrix} \begin{bmatrix} 57 \\ 19 \end{bmatrix}$$

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$$\underline{y} = \beta_0 + \beta_1 \underline{x} + \beta_2 \underline{x}^2$$

$$\begin{bmatrix} 1 \\ \underline{x} \\ \underline{x}^2 \end{bmatrix} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix}$$

$\underline{c}_1 \quad \underline{c}_2 \quad \underline{c}_3$

$$\underline{y} = \beta_0 \underline{c}_1 + \beta_1 \underline{c}_2 + \beta_2 \underline{c}_3$$

2-dimensional $\rightarrow$ 1 dimensional $\rightarrow$ prev example
(one independent)

Echap +

For symmetric Matrices: $A = A^T$

$$A = P D P^T$$

$\rightarrow$ Eigen-Decomposition / Diagonalization

$$Q^T Q = Q Q^T = I$$

$$A = P D P^{-1}$$

$= P D P^T$ reason for $P^T = P^{-1}$ always

Eigen-vectors of symmetric matrix are orthogonal

Quadratic forms:

A is symmetric

$$Q(x) = x^T A x, \quad x \in \mathbb{R}^n$$

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Example:

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$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= \begin{bmatrix} x_1 & x_2 \end{bmatrix}_{1 \times 2} \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}_{2 \times 2} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_{2 \times 1}$$

$$= \begin{bmatrix} x_1 & 3x_2 \end{bmatrix}_{1 \times 2} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_{2 \times 1}$$

$$= x_1^2 + 3x_2^2 \rightarrow \text{scalar}$$

$$\mathbb{R}^n \rightarrow \mathbb{R}$$

Theorem:

$\max (x^T A x) = \lambda_k$, where A is symmetric matrix & $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$ are eigen-values of A

subject to $x^T x = 1$

$$x^T u = 0$$

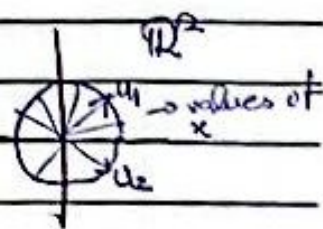
$$x^T u_1 = 0$$

$$x^T u_k = 0$$

$k=1$

$$\max (x^T A x) = \lambda_1, x = u_1$$

$$\text{next } \max (x^T A x) = \lambda_2, x = u_2$$



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$A = PDP^T$ is NOT possible for Non-sq matrices.

$$[A]_{m \times n} \rightarrow \text{SVD}$$

Symmetric/Square: $A = A$

$$Ax = \lambda x$$

x is an eigen-vec of A

$$\begin{aligned} \|Ax\| &= \|\lambda x\| \\ &= |\lambda| \|x\| \end{aligned}$$

x = unit vector

$$\|Ax\| = |\lambda|$$

$$x^T x = 1$$

Example,

$$A = \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix} \Rightarrow \underline{x} \rightarrow Ax$$

$$\|x\| = 1$$

Find a unit vector x at which length of Ax is maximized & compute the length

$$Ax \in \mathbb{R}^3$$

$$[x]_{3 \times 1}$$

$$b = Ax$$

$$\text{Arg Max}_{\underline{x}} \|Ax\| \Rightarrow \text{arg-max}_{\underline{x}} \|Ax\|^2$$

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$$\|Ax\|^2 = (Ax)'(Ax)$$

$$= \underbrace{x^T A^T A x}_{\text{symmetric}} \} \rightarrow \text{quadratic form}$$

$$(A^T A)^T = A^T A$$

=

$$\text{Max } \|x^T A^T A x\| \quad x^T x = 1$$

↓
eigen vector corresponding to
largest eigen value

$$A^T A \Rightarrow \begin{array}{l} \lambda_1 = 360 \rightarrow v_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \\ \lambda_2 = 90 \rightarrow v_2 = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} \\ \lambda_3 = 0 \rightarrow \frac{1}{3} \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} \end{array} \left. \vphantom{\begin{array}{l} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{array}} \right\} \text{orthogonal}$$

$$Av_1 = \begin{bmatrix} 18 \\ 6 \end{bmatrix}$$

$$\text{max } \|Ax\| = 4$$

$$\|Av_i\|^2 = v_i^T A^T A v_i$$

$$= v_i^T \lambda_i v_i$$

$$= \lambda_i \quad \text{if } \|v_i\| = 1$$

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$$\sigma_i = \sqrt{\lambda_i}$$

$$\sigma_i = \|Ax_i\|$$

$$[A]_{m \times n} = U \Sigma V^T \rightarrow \text{Right singular vector}$$

$m \times n$ $n \times n$

Diagonal
matrix having
singular vals of
 A .

↓
Eigen vectors
of $A A^T$
 $\{v_1, v_2, \dots, v_p\}$

Left singular
matrices
vectors

$$\begin{bmatrix} A v_1 & A v_2 & \dots & A v_n \\ u_1 & u_2 & & u_n \end{bmatrix}$$